ECE 6140 | HW01 | Brendon Simonsen

1. Passive and Active Sensors
   a) Alcohol thermometer **(Passive)**
   b) Thermostat in a car **(Passive)**
   c) pH meter in an aquarium
      i. **(Passive)** if it is just the chemical reaction with colors
      ii. **(Active)** if it is a digital pH meter
   d) Pressure sensor in the lid of a jar **(Passive)**
   e) Microphone in a cell phone **(Active)**
   f) Water level sensor in dishwasher **(Active)**
   g) Temperature sensor in a refrigerator **(Active)**
   h) Acceleration sensor in a car **(Active)**

2. Classification of sensors
   a) Both, this thermometer is measuring temperature. It is observing or reading a value. It then moves a dial to display the temperature.
   b) Passive
   c) The first transduction mechanism is the bimetal strip. It changes the energy from the heat, to expand the different metals, which could be considered kinetic energy. There is also a mechanism for taking the energy of expanding the metal and converting it to energy for moving the dial
3. Classification of a sensor
   i. This is an active sensor because it is sending out a stimulus to the hot wire with the expectation of a change to the wire temperature.
   ii. This is a relative sensor because it is using the relative power needed to keep the wire hot to determine the mass air flow.
   iii. Physical sensor because it uses the heat of the wire to determine if more current is needed to keep the wire hot. The current needed is then correlated with the mass air flow.
   iv. Thermoelectric. This sensor uses a correlation between heat, the temperature of the wire, and the voltage and current needed to maintain that temperature.
4. Acoustic pressure and dB.
   a) 120 dB
   b) Assuming that the attenuation is to avoid the threshold of pain, you would need 47.96 or ~48dB of attenuation. I would assume that for ear protection, you may err on the side of caution and use hearing protection with a greater amount of attenuation.
5. Simple RTD
   a) 54.7 m

b) 0.29 m

c) R(-45) = 89.58 Ω

R(120) = 166.8 Ω

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① 2×10^{-5} Pa (LOWEST) → 20 Pa (HIGHEST) Pa ⇔ dB $dB\cdot SPL = 20\log_{10}\left(\frac{P}{P_0}\right)$

(a) $20\log_{10}\left(\frac{2\times10^{-5}\,Pa}{2\times10^{-5}\,Pa}\right) = 0\,dB$ $(120-0) = 120\,dB$

$20\log_{10}\left(\frac{20\,Pa}{2\times10^{-5}}\right) = 20\log_{10}(10^6) = 120\,dB$

(b) 5,000 Pa ⇒ dB $20\log_{10}\left(\frac{5000}{2\times10^{-5}}\right) = 167.96\,dB$

$20\log(5000) - 20\log(20) = 20\log\left(\frac{5000}{20}\right) = 47.96\,dB \approx 48\,dB$

⑤ 0.1 mm THICK 120 Ω @ 20°C $A_{circle} = \pi r^2$

a) $R = \frac{L}{\sigma A}$ ⇔ $L = R\sigma A$ $L = 120(5.8\times10^7)\left(\pi\left(\frac{0.1E-3}{2}\right)^2\right) = 54.7\,m$

$R(t) = R_0(1+\alpha[T-T_0])$

b) $C = \pi D = \pi(0.006) = 0.01885\,m$ $\frac{L_{WIRE}}{C} = \frac{54.7\,m}{0.01885\,m} = 2.9E3$ TURNS

WIRE THICKNESS = 0.0001 m

$(0.0001)(2.9E3) = 290\,mm \Rightarrow 0.29\,m$
{m} {TURNS}

c) -45°C +120°C

$R(t) = R_0(1+\alpha[T-T_0])$

$\alpha = 0.0039$

$T_0 = 20°C$

$R_0 = 120\,\Omega$

$R(-45°C) = 120(1+0.0039[-45-20]) = 89.58\,\Omega$

$R(+120°) = 120(1+0.0039[120-20]) = 166.8\,\Omega$

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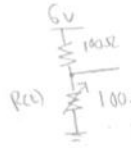
6) -200°C to +600°C @ 0°C R = 100 Ω

TCR = 0.00385/°C

SELF HEAT = 0.07 °C/mW

POWER SOURCE

6V 100Ω RESISTOR



$R(t) = R_0(1 + \alpha[T - T_0])$

$I(0) = \frac{V_s}{R} = \frac{6V}{(0.03+100)\Omega} = 0.03A$

$P = I^2 R(t)$

$P = (0.03A)^2(100\Omega) = 0.09W$

100°C $SH = (0.07\,°C/mW)(P_{mW})$

$P_{SH} = (0.07\,°C/mW)(90E-3) = \boxed{6.3°C}$

$R(100) = 100(1 + 0.00385[100-0]) = 138.5\Omega$

$I(100) = \frac{V}{R} = \frac{6V}{(100+138.5)} = 25.16E-3A$ $P = (25.16E-3A)^2(138.5) = 87.65E-3W$

$SH = (0.07)(87.65mW) = \boxed{6.1355°C \text{ ERR}}$

80°C $R(80) = 100(1 + 0.00385[80-0]) = 130.8\Omega$ $I(80°) = \frac{6V}{(100+130.8)} = 25.94 \approx 26E-3A$

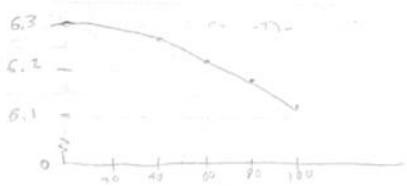
$P = (26E-3)^2(130.8) = 88.4E-3W$ $SH = (0.07)(88.4) = 6.19°C$ ERR

60°C $R(60) = 100(1 + 0.00385[60-0]) = 123.1\Omega$ $I(60°) = \frac{6V}{(100+123.1)} = 26.9E-3A$

$P = (26.9E-3)^2(123.1) = 89.04E-3W$ $SH = (0.07)(89.04) = 6.2328°C$ ERR

40°C $R(40) = 100(1 + 0.00385[40-0]) = 115.4\Omega$ $I(40) = \frac{6V}{100+115.4} = 27.86E-3A$

$P = (27.86E-3A)^2(115.4\Omega) = 89.5E-3W$ $SH = (0.07)(89.5) = 6.265°C$ ERR



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(7) $R(t) = R_0(1 + \alpha[T - T_0])$ $\alpha = 1.8 \times 10^{-3}$ TCR = 0.0056

22Ω @ 20°C 120V, 100W LIGHT BULB

a) TEMP OF FILAMENT WHEN BULB IS LIT $P = \frac{V^2}{R} \Leftrightarrow R = \frac{V^2}{P}$

$R(t) = \frac{120^2}{100} = 144$

$\left(\frac{\frac{R(t)}{R_0} - 1}{\alpha}\right) + T_0 \Rightarrow \left(\frac{\frac{144}{22\Omega} - 1}{0.0056}\right) + 20 = \boxed{1.01E3\,°C}$

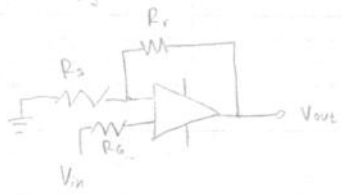
b) ONE SOURCE OF ERROR COULD BE THAT THE SURFACE AREA OF THE FILAMENT MAY VARY WHICH COULD CHANGE THE RESISTANCE OF THE WIRE.

THE ENVIRONMENT MAY NOT ALWAYS BE 20°C. OUR CALCULATIONS USE THE RESISTANCE OF THE WIRE AT 20°C.

(8) RANGE → 0 AND 100 μV OUTPUT 0V → 5V $Z_{IN} = 10M\Omega$ OPEN LOOP $G = 10^6$

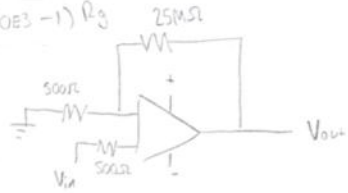
$V_{out} = 0 \to 5V$ $V_{in} = 0 - 100\,\mu V$ $G = \frac{5V}{100\mu V} = 50E3$

$G = 1 + \frac{R_f}{R_g}$ $R_s = R_g$ $\boxed{R_g = 500\Omega,\ R_F = 25M\Omega}$



$R_F = (G-1)R_g \Rightarrow R_F = (50E3 - 1)R_g$

R_g	R_f
100K	~5E9
200K	~10E9
100	~5M
500	~25M



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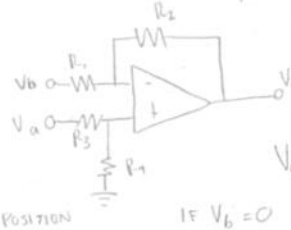
(9) FOR $V_{out} = V_a - V_b$ IF $\underbrace{R_1 = R_2 = R_3 = R_4}_{1}$ $V_a = 0$

IF $R_1 = R_2 = R_3 = R_4$ THEN



V_{OUT} EQUATION FOR SUBTRACTING AMPLIFIER:

$$V_{out} = -V_b\left(\frac{R_2}{R_1}\right) + V_a\left(\frac{R_4}{R_3+R_4}\right)\left(\frac{R_1+R_2}{R_1}\right)$$

USING SUPERPOSITION

$V_{out} = V_a - V_b$

IF $V_b = 0$ AND $R_1 = R_2 = R_3 = R_4$

$V_{out} = V_a$

IF $V_a = 0$ AND $R_1 = R_2 = R_3 = R_4$

$V_{out} = -V_b$

b) $V_{out} = 5(V_a - V_b)$

IF $R_2 = 5$ AND $R_1 = 1$ $\quad V_{out} = -V_b(5) + V_a\left(\frac{R_4}{R_3+R_4}\right)\left(\frac{1+5}{1}\right)$

$5 = \left(\frac{R_4}{R_3+R_4}\right)\left(\frac{6}{1}\right) \Rightarrow \frac{5}{6} = \left(\frac{R_4}{R_3+R_4}\right)$ $\quad R_4 = 5$ AND $R_3 = 1$ $\quad \frac{5}{1+5} = \frac{5}{6}$

$\boxed{\text{IF } R_1 = 1\Omega,\ R_2 = 5\Omega,\ R_3 = 1\Omega,\ R_4 = 5\Omega}$

(10) $1k\Omega$ @ $25^\circ C$ MATERIAL CONSTANT $\beta = 3{,}200\,K$ 0.2 mA CURRENT SOURCE

TEMP RANGE $0^\circ C - 50^\circ C$ $\quad 25^\circ C = 298.15K$ $\quad R(T) = R_0 \cdot e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}$

a) CALC ERROR IN TEMP $\quad 50^\circ C = 323.15K$ $\quad R_0 = 1k\Omega$ $\quad T = 323.15K$ $\quad T_0 = 298.15K$

$R(50^\circ C) = R(323.15K) = 1k\Omega \cdot e^{3200K\left(\frac{1}{323.15} - \frac{1}{298.15}\right)} \Rightarrow R(50^\circ C) = 435.9\Omega$

$$T = \frac{\beta}{\ln\left(\frac{R(T)}{R_0 e^{-\beta/T_0}}\right)} = \frac{3200}{\ln\left(\frac{435.9}{1000 \cdot e^{-3200/298.15}}\right)} = 323.09K \Leftrightarrow 49.94^\circ C$$

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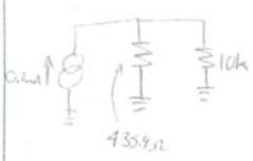
11.1)



$$\frac{10k}{435.9\Omega + 10k} = 0.9582 \left(\frac{V}{V}\right)$$

$$1 - 0.9582 = 0.418 = 4.2\%$$

a) 4.2% ERROR IN TEMPERATURE INTRODUCED BY THE INPUT RESISTANCE WHEN AT 50°C

b) I EXPECT THE RANGE TO BE LARGEST WHEN THE TEMPERATURE IS THE COLDEST. THIS IS BECAUSE R_S INCREASES AS TEMPERATURE DECREASES.

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(11)

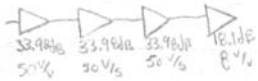
$G = 120\ dB$ INDIVIDUAL AMPLIFIER RANGE 1-50

$G_{dB} = 20\log(G)$ $\quad 10^{\left(\frac{120}{20}\right)} = 1E6$

$G = 10^{\left(\frac{G_{dB}}{20}\right)}$

$G_{dB} = 20\log(50) \approx 34\,dB = 33.979$ $\quad \frac{120\,dB}{33.979} = 3.53.$

SO THIS AMPLIFICATION OF 120dB COULD BE DONE IN 4 AMPLIFIERS



$120-(3\cdot 33.979) = 18.063 \approx 18.1\,dB$

$10^{\left(\frac{18.1}{20}\right)} = 8.04 \approx 8$

THIS SOLUTION IS NOT UNIQUE. ANY COMBINATION OF AMPLIFIERS THAT, IN dB, ADD TO 120dB OF GAIN.